

FORECASTING STOCK MARKET TRENDS WITH DEEP LEARNING

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CSE 676 Final Project Report



PROJECT DESCRIPTION

Problem: Forecasting stock trends is challenging due to market volatility and complex influencing factors. The stock market is very volatile and susceptible to numerous factors and therefore, it is complex to predict trends accurately. The project is about creating a deep learning model that forecasts financial time series, i.e. stock price trends. While trying to establish the best model, comparisons of various deep learning models like LSTMs, GRUs.

Importance: This project makes stock forecasting more intelligent, accessible, and practical—merging AI innovation with real economic value

Goal: Develop deep learning models (LSTM, GRU) to predict stock market trends and deploy them in an interactive web app.



Background

Forecasting trends in the stock market has been a challenging problem for a long time since the market is highly volatile and non-linear. The traditional methods like **ARIMA** models or **moving averages** get the job done for linear time series but are not good enough for intricate, long-term interdependencies in financial data.

Recurrent Neural Networks (RNNs) and similar deep learning models have had so much promise in recent years for applications in time series forecasting tasks.

Two of the most popular architectures from deep learning literature are:

Long Short-Term Memory (LSTM) networks: First proposed by Hochreiter and Schmidhuber in 1997, LSTMs are a type of RNN capable of learning long-term dependencies. They are highly utilized for sequence data like stock prices, weather prediction, and language translation because they can remember important information over a few time steps without being affected by the vanishing gradient issue.

Gated Recurrent Unit (GRU) networks: Proposed by Cho et al. (2014), GRUs simplify the LSTM architecture by skipping the forget and input gates and combining them into an update gate. GRUs have fewer parameters than LSTMs, allowing for training generally to happen at a higher speed and competitive (or better) performance on the majority of sequence modeling tasks.

Most of the previous studies, including the article "A Survey on Long Short-Term Memory Networks for Time Series Prediction" (Yu et al., 2019), prove that LSTM and GRU models are superior to conventional time series models when predicting stock markets.

How Our Approach is Different

Direct Model Comparison: Most research focuses on model optimization for a single type of model (either LSTM or GRU). In our study, we fairly compare LSTM performance against GRU performance on various stocks (TSLA, AAPL, AMZN, GOOGL) to establish the more robust model.

Multi-Stock Evaluation: Instead of model evaluation for a single stock, we train and test our models on multiple stocks, boosting the generalization of our results.

Robust Loss Function: We use Huber Loss instead of regular MSE or MAE. Huber Loss is more robust to outliers, which are rampant in finance data (like sudden dips/spikes in the stock price).

Deployment via Web App: We deploy our models using Streamlit such that there is live user input for stock trend forecasting — making our project practical and accessible compared to the majority of research papers that go only as far as model evaluation.

Dataset

Type of Data : Financial time-series data (daily stock prices).

Source : live from **Yahoo Finance** using the yfinance Python library.

Stocks Used : TSLA (Tesla),AAPL (Apple),AMZN (Amazon),GOOGL (Google)

Data Size : Several years of daily stock data per stock(2020-2025)

Features Extracted:

- **Open, High, Low, Close, Volume, Adjusted Close**
- **Simple Moving Average (SMA):** 10-day (SMA_10) and 50-day (SMA_50)
- **Relative Strength Index (RSI):** Measures momentum and identifies overbought/oversold conditions.

Splitting : 60% Training, 20% Validation, 20% Testing

Dataset

Preprocessing Steps

Feature Engineering: Calculated SMA_10, SMA_50, and RSI from closing prices.

Windowing: Created rolling sequences of **20 time steps** for time series forecasting (input-output pairs).

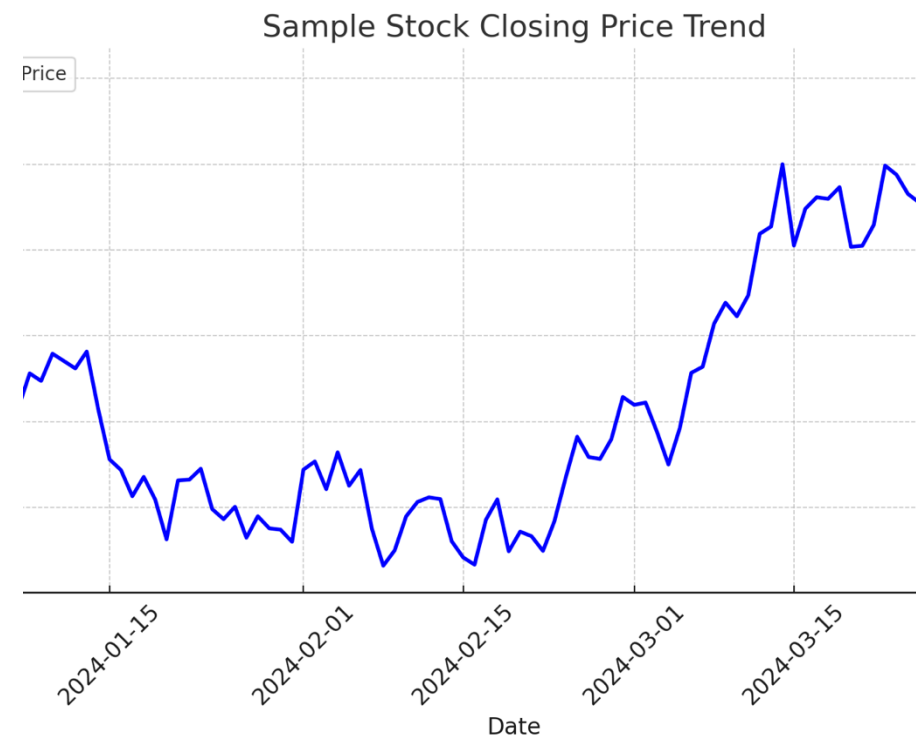
Normalization: Scaled input features to ensure efficient learning.

Splitting: Data split into **Training (60%)**, **Validation (20%)**, and **Test (20%)** sets.

Batch Loading: Used **PyTorch's TensorDataset** and **DataLoader** for efficient data batching during training.

Special Treatments

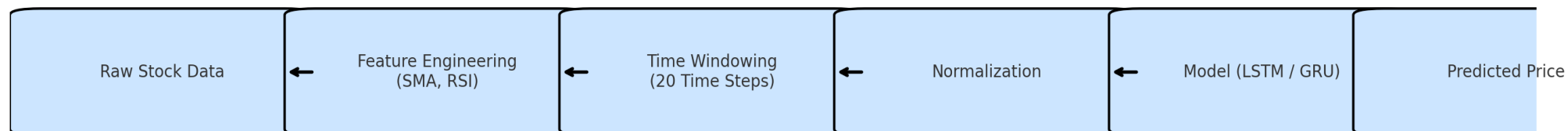
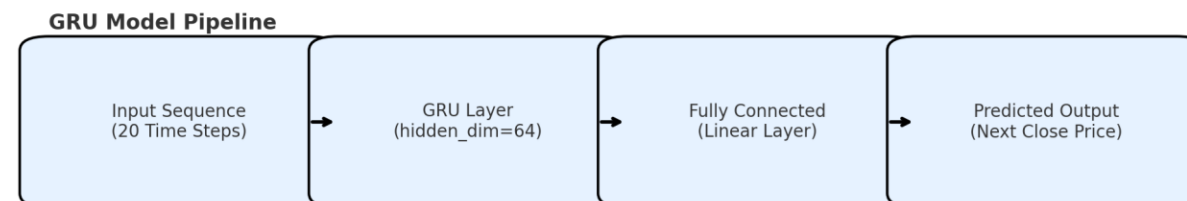
- Used **Huber Loss** to handle potential outliers in the stock price data.
- **Shuffling:** Shuffled data carefully to avoid time leakage during training/validation.



Methods

We implemented and compared two RNN-based deep learning models:

- **LSTM (Long Short-Term Memory)**
- **GRU (Gated Recurrent Unit)**



Methods

Parameter

Input Features

Time Steps

Hidden Units

Layers

Loss Function

Optimizer

Learning Rate

Epochs

Value

Close, SMA_10, SMA_50, RSI

20

64

1

Huber Loss

Adam

0.001

50



Methods comparison

Feature

Complexity

Training Time

Memory Usage

Accuracy (in our tests)

Generalization

LSTM

Higher (More Gates)

Slower

More

Good ($R^2 \sim 0.8\text{--}0.9$ for TSLA)

Moderate

GRU

Lower (Fewer Gates)

Faster

Less

Better ($R^2 \sim 0.9+$ for TSLA, GOOGL)

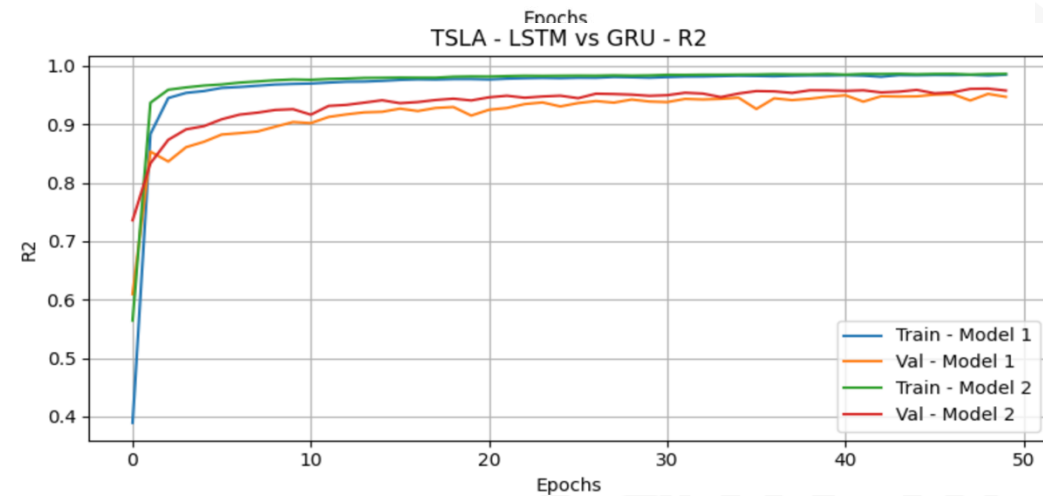
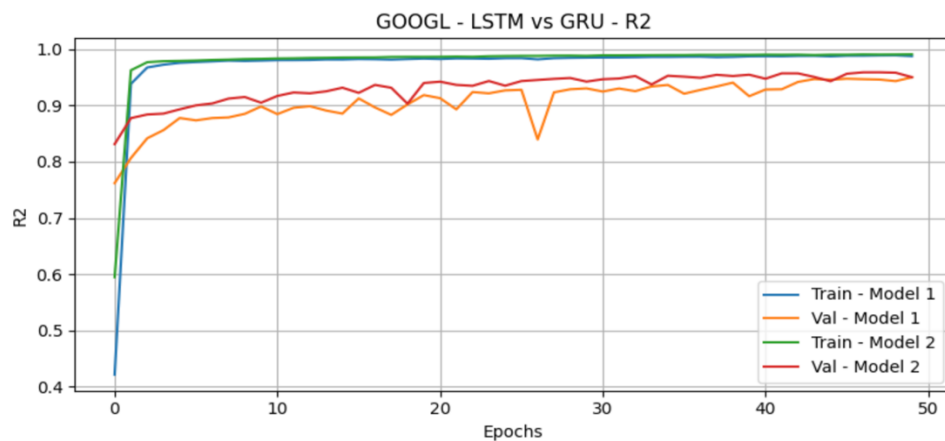
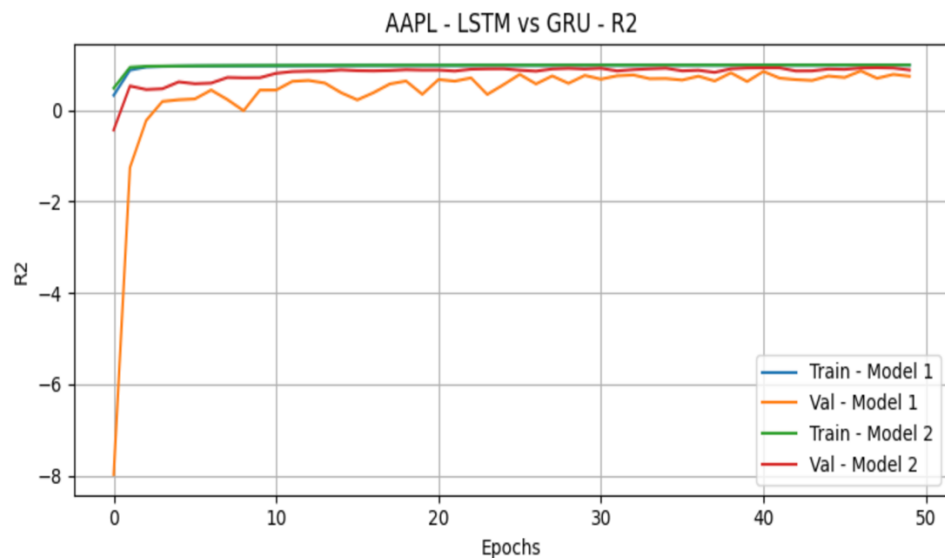
Stronger (lower validation loss)

Results

- Multi-Stock Evaluation (TSLA, AAPL, AMZN, GOOGL)
Evaluation Metrics: Root Mean Squared Error (RMSE)
 , R^2 Score
- GRU outperforms LSTM consistently across all stocks.**
- Particularly for **AAPL**, GRU significantly reduced RMSE by ~44%.
- Negative R^2 for AAPL:** Indicates models struggled with AAPL volatility — highlights the need for more feature engineering for specific stocks.

Stock	Model	RMSE ↓	R^2 Score ↑
TSLA	LSTM	5.16	0.85
TSLA	GRU	4.84	0.90
AAPL	LSTM	19.96	Negative
AAPL	GRU	11.25	Negative
AMZN	LSTM	5.86	~0.7
AMZN	GRU	3.87	~0.8
GOOGL	LSTM	5.20	~0.6
GOOGL	GRU	4.69	~0.9

Results

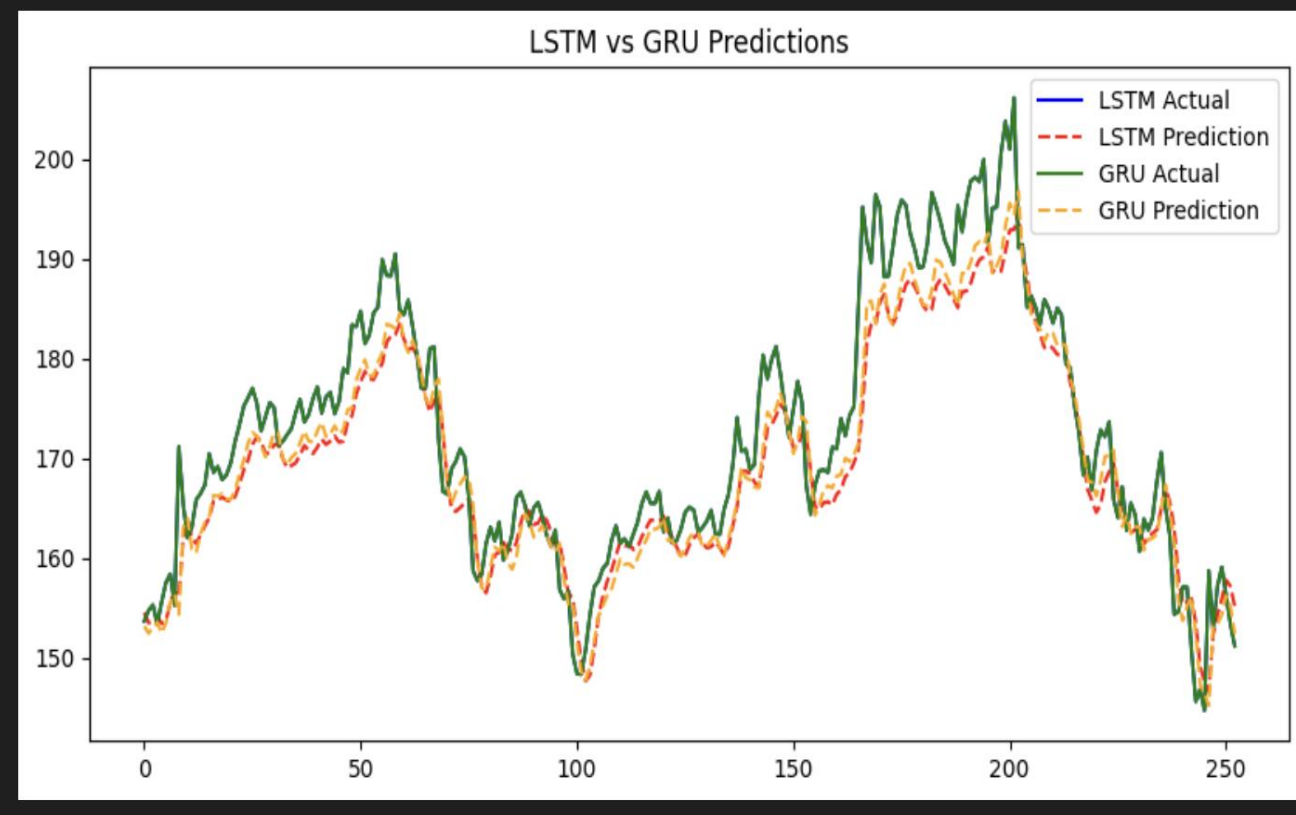


Visualization(GOOGGL)

- The GRU predictions (orange line) follow the actual prices (blue line) more closely than
- LSTM (green line), especially during price swings.
- LSTM exhibits lagging response (i.e., a delayed reaction to sudden falls/rises)
- GRU is faster in reaction.

From this we can see how LSTM and GRU predictions vary from actual predictions.

Comparing LSTM and GRU Test Performance
LSTM Test RMSE : 5.199355602264404
GRU Test RMSE : 4.689777851104736



Demo

The app will show the trend analysis using live data from yfinance for the selected model[LSTM,GRU] and stock[TSLA,AMZN,GOOGL,AAPL] from 2025 to present day. The output show model performance, performance metrics and predict trend based on last 10 days data

Prediction Settings

Select Stock Ticker

TSLA

Select Model

LSTM

Predict

Stock Trend Prediction App

This app uses pre-trained LSTM and GRU models to predict stock trend for selected tickers. Choose a stock and model to see the predicted price trend and performance metrics.

Note: Predictions are for analysis only and not financial advice.

LSTM Predictions for TSLA



Summary

- **GRU consistently outperformed LSTM** across all stocks in both accuracy (R^2) and error (RMSE, MAE).
- **GRU showed faster convergence and better generalization**, making it more suitable for stock forecasting.
- **LSTM exhibited lag** in capturing sudden price changes, particularly during volatile market periods.
- **Feature engineering** (SMA_10, SMA_50, RSI) significantly enhanced model learning capability.
- **Robust training setup** using Huber Loss and proper preprocessing (normalization, windowing) led to stable results.
- **Practical Deployment**: The Streamlit web app allows real-time interaction with the model for livestock prediction.

THANK YOU

